

Metaheuristic Algorithms and Applications: Program Report II-2

Program Report on Clustering and Classification

Deadline: May 25, 2026

Apply the metaheuristic algorithms based clustering and classification approaches to the **Early Stage Diabetes Risk Prediction**, which is referred to UCI Machine Learning Repository [<http://archive.ics.uci.edu/ml/index.php>]. Irvine, CA: University of California, Center for Machine Learning and Intelligent Systems.

(<https://archive.ics.uci.edu/dataset/529/early+stage+diabetes+risk+prediction+dataset>)

Abstract: This dataset contains the sign and symptom data of newly diabetic or would be diabetic patient.

Data Set Characteristics:	Multivariate	Number of Instances:	520	Subject Area:	Computer Science
Attribute Characteristics:	Categorical, Integer	Number of Attributes:	16	Date Donated	07-11-2020
Associated Tasks:	Classification	Missing Values?	N/A	Number of Web Hits:	49211

Data Set Information:

This has been collected using direct questionnaires from the patients of Sylhet Diabetes Hospital in Sylhet, Bangladesh and approved by a doctor.

Attribute Information:

Age 1. 20-65; **Sex** 1. Male, 2. Female; **Polyuria** 1.Yes, 2.No; **Polydipsia** 1.Yes, 2.No.
sudden weight loss 1.Yes, 2.No.; **weakness** 1.Yes, 2.No.; **Polyphagia** 1.Yes, 2.No.
Genital thrush 1.Yes, 2.No.; **visual blurring** 1.Yes, 2.No.; **Itching** 1.Yes, 2.No.
Irritability 1.Yes, 2.No.; **delayed healing** 1.Yes, 2.No.; **partial paresis** 1.Yes, 2.No.
muscle stiffness 1.Yes, 2.No.; **Alopecia** 1.Yes, 2.No.; **Obesity** 1.Yes, 2.No.
Class 1.Positive, 2.Negative.

Citation Request:

Likelihood Prediction of Diabetes at Early Stage Using Data Mining Techniques

By M. M. F. Islam, Rahatara Ferdousi, Sadikur Rahman, Humayra Yasmin Bushra. 2019. Published in Computer Vision and Machine Intelligence in Medical Image Analysis

<https://www.semanticscholar.org/paper/Likelihood-Prediction-of-Diabetes-at-Early-Stage-Islam-Ferdousi/9329dec57c5f13f195220ffa7077fd0029983f07>

Program Report should include the following two parts.

Part I: Perform feature (attribute) extraction, clustering, and classification to make a correct diagnostic (prediction), using the same data for both training and testing.. Also, evaluate the corresponding Accuracy, Sensitivity (Recall or True positive rate), Specificity (True negative rate), Precision, and F-measure.

Part II: Perform 10 repetitions of 5-fold cross-validation for feature (attribute) extraction, clustering, and classification. Report the mean and standard deviation of the accuracy rates for both the training and testing data.

Questions:

Do you have to use all the attributes?

Is it possible to find the best combination of attributes to make the highest classification Accuracy rates?